

The current remains constant as the rate of change of flux is constant.

- 27** When a light bulb is connected across an a.c. source of peak voltage 170 V, the power dissipated is 40 W. Two such light bulbs are now connected in series to the electrical mains of 220 V r.m.s. What will be the total power dissipated?

A 33.5 W

B 67.0 W

C 80.0 W

D 134.0 W

Ans: B

$$V_{rms} = \frac{V_o}{\sqrt{2}}$$

$$\langle P \rangle = \frac{V_{rms}^2}{R}$$

$$40 = \frac{\left(\frac{170}{\sqrt{2}}\right)^2}{R}$$

$$R = 361.25$$

$$\langle P_{Total} \rangle = \frac{V_{Total,rms}^2}{R} = \frac{(220)^2}{361.25 \times 2} = 67.0 \text{ W}$$

20. Light of wavelength λ is incident on a metal surface in a vacuum. Photoelectrons are emitted